dbGap Task 2 Cohort Feasibility and Study Discovery

Task 2 participants are tasked with developing an intuitive, privacy-preserving, reasoning-enabled, and conversational AI interface for cohort feasibility assessment and study discovery. Systems should enable investigators to formulate complex research information needs in natural language and reason over the available evidence in the challenge dataset -including publicly available dbGaP study documentation, data dictionaries, genotype and genomic metadata, and aggregated or summary statistics - to identify studies relevant to those needs.

Relevance should be determined by the required combination of study characteristics specified in the information need, rather than by the presence of individual terms or concepts in the study documentation. Systems should therefore demonstrate the ability to perform multi-hop reasoning across study characteristics, integrating multiple pieces of evidence, resolving relationships among characteristics, distinguishing required characteristics from incidental ones, and providing a clear, reasoned basis for why a study meets the investigator's stated criteria.

The following provides an example information need and the corresponding expected response for this Challenge Task:

Information need: Find dbGaP studies involving adult participants with lung adenocarcinoma as the primary disease, where participant-level data include a history of smoking or smoking exposure and whole genomic sequences.

Expected Response:

List of Relevant Study Accessions:

- "phs001954.v2.p1",
- …
- other study accessions considered relevant to this information need.

In this example, a study is considered relevant only if the following information needs are satisfied for the example provided above:

- Adult participants — the study includes adult participants.
- Primary disease — lung adenocarcinoma is the study's primary disease or condition of interest.
- Smoking history or exposure — participant-level data include information on smoking history or smoking exposure.
- Whole-genome sequencing — participant-level data include whole-genome sequencing (WGS) data.

A study that fails to satisfy any one of these needs will be deemed not relevant.

For the returned study phs001954.v2.p1, here are some details on how the study meets the information need in terms of extracted excerpts from study related files found the challenge corpus:

Study: phs001954.v2.p1
Study Name: *International Cancer Proteogenome Consortium (ICPC): Proteogenomics of Early Stage Lung Adenocarcinoma in Taiwan*

Need	Supporting Evidence	Source
1. Adult participants	The study's `AGE` variable is measured in years, with a documented range of **32–85 years**, establishing that the study can be inferred to include adult participants.	`pht009798.v2.p1.data_dict.xml`
2. Lung adenocarcinoma as the primary disease	The study metadata identifies the `main_disease` as **"Adenocarcinoma of Lung."**	`phs001954.v2.p1.json`
3. Smoking history or exposure	Study documentation/data dictionaries contain variables documenting **smoking history or smoking exposure**.	Appropriate dbGaP study/data-dictionary file(s)
4. Whole-genome sequencing	The dataset metadata identifies **WGS** among the available study data types, along with RNA-Seq and WXS.	`pht009796.v2.p1.json`

Challenge Dataset

Participants will be provided with a fixed snapshot of dbGaP documents, associated metadata, and data dictionaries, including relevant version information where applicable. Responses to Task 2 information needs must be limited to and based on this snapshot. The snapshot is derived from official NCBI public repositories and includes the document content and associated metadata required to support the challenge tasks. Participants will therefore work from a common, fixed corpus, rather than independently downloading or accessing documents from dbGaP. The challenge dataset does not require access to controlled-access individual-level data. ***Use of controlled-access individual-level data is strictly prohibited. Any submission found to have used controlled-access individual-level data, either directly or indirectly through an external data source, service, or system, will be disqualified from the challenge.***

The data for this challenge can be downloaded via the following link:

- https://ftp.ncbi.nlm.nih.gov/dbgap/NLM_Challenge/dbgap.tar.gz

This will result in one compressed TAR file being download, dbgap.tar.gz. Participants should unpack the archive to create a base directory named dbgap. The directory contains a README.txt file introducing the snapshot of dbGaP (the database of Genotypes and Phenotypes) studies and their associated objects, together with supporting reference material.

The dataset is organized so that each study and its associated objects can be inspected independently while preserving the relationships between objects:

```
dbgap/
├── studies/          one directory per dbGaP study, containing its objects
├── collections/      dbGaP collections (logical groupings of studies)
├── collections.json  aggregate summary of collections and their members
└── web/              supporting dbGaP reference pages (HTML/PDF)
```

At a glance, the Challenge Dataset covers:

Object	dbGaP Prefix	Approx. Count
Studies	phs	3,578
Phenotype datasets	pht	16,070
Variables	phv	465,853
Genotype datasets	phg	2,122
Documents	phd	8,079
Analyses	pha	17,277
Collections	phs (collection hosts)	23

The total size of the snapshot is approximately 6.3 GB unpacked (1.6 GB Compressed), dominated by approximately 466,000 variable JSON files and per-object HTML captures.

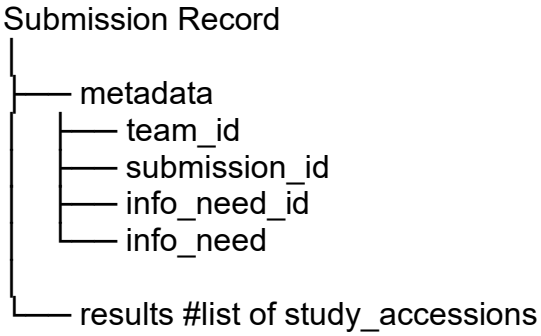
Task 2 Submission JSON Structure

On January 31, 2027, Participants in this challenge will submit a submission file containing responses to up to 1,000 natural-language information needs comprising the evaluation set. On January 15, 2027, Participants must also submit a Docker-based system, a description of their system, including its components, technical documentation, and deployment and execution instructions, in accordance with the System Submission requirements provided below.

Each response submission file should be a UTF-8-encoded JSONL file, with each line being a JSON object for an information need. The participants can submit up to three (3) submissions for the task. Participants should follow this format and upload their submission files via the Challenge portal.

Participants will be provided the evaluation set of information needs on January 18, 2027. Subsequently, participants should develop responses to these needs and upload the submission file via the challenge portal prior to January 31, 2027. Submissions file should be UTF-8-encoded and use JSONL file structure, with each line being a JSON object for an information need. The metadata object identifies the participating team, system run, and information need and contains the original request. The response object contains the system's retrieval results. The records array contains one or more retrieved studies, with each study being identified by an accession number.

The Task 2 submission file is a UTF-8-encoded JSONL (JSON Lines) file. Each JSONL record represents the system's response to one information need. At a high-level, the Submission Record has the following hierarchical structure:



Field Definitions

Level	Field	Description
Record	`metadata`	Identifies the submission and information need

Level	Field	Description
	`results`	Array containing the studies returned by the system
Metadata	`team_id`	Unique identifier for the participant team
	`submission_id`	Identifier for the specific system submission/run
	`info_need_id`	Unique identifier for the information need
	`info_need`	Original natural-language information need
Results	`Ordered list of study_accessions`	Rank/order of the returned dbGaP study accessions

Example

The following is an example of the submission record:

```
{
  "metadata": {
    "team_id": "team-001",
    "submission_id": "team-001-run-01",
    "info_need_id": "dbT2_1",
    "info_need": "Find dbGaP studies involving adult participants with lung
adenocarcinoma as the primary disease, where participant-level data include a
history of smoking or smoking exposure and whole-genome sequencing data."
  },

  "results": ["phs001954.v2.p1",....]
}
```

System Submission

By January 15, 2027, participants must submit a Docker-based system, a description of their system and its components, together with sufficient technical documentation and instructions to enable NIH to reproduce, deploy, and evaluate the system within an NIH-managed AWS environment. All software, models, indexes, data resources, documentation, and other materials that comprise or support the submitted system must have been created or last modified on or before January 15, 2027. A link to the participant's GitHub/Docker repository for the solution will be accepted.

Submissions should describe the system architecture and components, including retrieval, ranking, embeddings, indexes, models, LLM inference, and any other software or data resources required to execute the system. Participants may design their systems as a single containerized application or as multiple containerized services, including distributed architectures in which different components run on separate compute nodes. Participants should also provide deployment and execution instructions, along with an estimate of the computational and storage resources required (e.g., CPU, memory, GPU, and storage).

For evaluation, NIH will provision and manage the underlying AWS infrastructure, including compute, storage, networking, and orchestration. Participants will not be required to provision or manage AWS resources, Kubernetes clusters, or Terraform infrastructure.

To support verification of the January 15, 2027 submission deadline, participants may be required to provide cryptographic hashes or other verifiable records for software, models, indexes, data artifacts, and other system components identified in their submission. When requested during the evaluation phase, the corresponding executable or data artifacts must match the versions documented and verified at the submission deadline.

Model and Resource Restrictions: Submitted systems may use only publicly available, open-weight Large Language Models (LLMs), open-source software, and/or other publicly available computing resources. Systems may not invoke external APIs to proprietary frontier models (e.g., commercial hosted LLM endpoints) or rely on non-public, restricted-access software, models, datasets, or other resources during execution. Any pretrained/fine-tuned models used by the system must be included within the submitted Docker images.

- Docker Image Execution and Network Restrictions
  - The challenge organizers reserve the right to run the submitted Docker-based system and verify system reproducibility, resource utilization, and compliance with challenge rules. Specifically:
  - External Calls: Submitted Docker Images will be executed in a completely isolated environment with all external network calls blocked. Limited access will be permitted to NLM-hosted services, such as the UMLS Terminology Services (UTS), and other challenge specific services. Details of these approved services and endpoints will be made available when the challenge is published. This zero-egress policy (excluding the permitted services) prevents both data leaks and runtime reliance on external computing resources, tools, or services.

Additional technical guidance, including submission formats, deployment specifications, resource limits, verification requirements, and evaluation procedures, will be provided to participants sufficiently in advance of the January 15, 2027 submission deadline.

Evaluation and Challenge Winner Selection

To determine the challenge winners for this task, participants will submit their developed systems and a submission file containing responses to up to 1,000 natural-language task specific information needs comprising the evaluation set. Evaluation will be conducted in two phases.

Phase 1: System Selection

The evaluation dataset will be a selected subset of 200 information needs for the initial system-selection phase, which will identify the N top-performing systems for more extensive evaluation in Phase 2. For this evaluation set, all participant submissions will be evaluated against expert-verified reference set of relevant studies. The primary evaluation metric will be Normalized Discounted Cumulative Gain (NDCG), which measures both the relevance of retrieved studies and their ranking, giving greater weight to highly relevant studies appearing near the top of the ranked list.

A higher NDCG indicates better-performing systems.

The top N (expected to be five) systems based on Phase 1 NDCG performance will advance to Phase 2.

Phase 2: Extended Evaluation and System Deployment

In this phase, each of the top N systems will be installed and configured in an NIH-provided AWS cloud environment to ensure a standardized evaluation environment. These systems will be evaluated using the private set of information needs, ease of system installation, resource usage (memory, disk, compute), and response time.

Phase 2 will use the following evaluation criteria:

- **NDCG**: Measures the relevance and ranking quality of retrieved studies against the expert-verified reference set.
- Ease of Use: Assesses the effort required to deploy, configure, operate, and use the system in the standardized NIH cloud environment.
- Compute Time: Measures the computational time required to generate responses.
- Compute Effort: Measures the computational effort required to generate responses, including the resources and processing required by the system.

- Memory Footprint: Measures the memory and other computational resources required to operate the system and generate responses.

NDCG will serve as the primary metric for determining final system rankings and identifying the overall winning systems. In the event of a tie in this metric, the additional evaluation criteria listed above may be used to differentiate systems and determine the final ranking.